AI PRODUCT MANAGEMENT 2 · DETAILED COURSE

Retrieval-Augmented Generation (RAG)

Grounding model answers in your own data: architecture, retrieval quality, evaluation and production operation.

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CHAPTER 14

Retrieval-Augmented Generation (RAG)

Grounding model answers in your own data: architecture, retrieval quality, evaluation and production operation.

RAG Fundamentals

Retrieve relevant documents at request time and give them to the model to ground its answer.

Retrieval-augmented generation finds relevant documents at request time and gives them to the model so its answer is grounded in them. It lets a product answer from current company data without retraining the model.

Why RAG Is Required

Models lack your private or recent data and can hallucinate; RAG supplies sources and enables citations.

Models do not know your private or recent data and can hallucinate. RAG supplies sources, which enables citations and makes answers checkable.

RAG vs Fine-Tuning

RAG adds knowledge at query time and is easy to update; fine-tuning changes behaviour or style. They are often combined.

RAG adds knowledge at query time and is easy to update. Fine-tuning changes behaviour, style or format. Many products start with RAG and fine-tune only if behaviour still falls short.

RAG Architecture

Ingest and index, retrieve, rerank, build context, generate, cite, and monitor.

The pipeline ingests and indexes documents, retrieves candidates for a query, reranks them, builds the context, generates the answer with citations and is monitored for quality.

Data Ingestion

Collecting documents from sources and keeping the index in sync with changes.

Ingestion collects documents from sources such as drives, wikis and databases, then keeps the index in sync when content changes. Stale content is a common cause of wrong answers.

Document Parsing

Extracting clean text, tables and structure from PDFs, web pages and office files.

Parsing extracts clean text, tables and headings from PDFs, web pages and office files. Poor parsing silently degrades every later step, so check parsed output by hand.

Data Cleaning

Removing noise, duplicates and boilerplate and fixing encoding so retrieval is accurate.

Cleaning removes boilerplate, duplicates and noise, and fixes encoding problems. Clean data makes retrieval more accurate and smaller.

Document Chunking

Splitting documents into passages sized for retrieval and context limits.

Chunking splits documents into passages sized for retrieval and the context window. Chunks that are too small lose meaning; chunks that are too large dilute relevance.

Chunking Strategies

Fixed size, recursive, by heading or semantic boundaries, with overlap to preserve context.

Options include fixed-size chunks, recursive splitting, splitting by heading, and semantic boundaries, often with some overlap to keep context. Evaluate each option on your own questions.

Embedding Models

Models that convert text to vectors; choose by quality, language coverage, dimensions and cost.

Embedding models turn text into vectors. Compare them on retrieval quality for your language and domain, plus dimensions, cost and speed.

Vector Databases

Stores optimised for nearest-neighbour search over embeddings.

Vector databases store embeddings and run fast nearest-neighbour search. They also support metadata filters. Choose based on scale, hosting needs and integration.

Pinecone

A managed vector database service with simple scaling.

Pinecone is a managed vector database service with simple scaling. Teams that do not want to run infrastructure often choose a managed option.

Weaviate

An open-source vector database with hybrid search and modules.

Weaviate is an open-source vector database with hybrid search and modules for embedding. It can be self-hosted or used as a managed service.

Chroma

A lightweight open-source vector store popular for prototypes.

Chroma is a lightweight open-source vector store, popular for prototypes and small applications. Check its production readiness for your scale before committing.

FAISS

Meta's library for efficient similarity search over vectors, run in your own process.

FAISS is a library from Meta for efficient similarity search, run inside your own process. It is fast but does not provide a full database service by itself.

Semantic Search

Finding passages by meaning using embeddings, even without shared keywords.

Semantic search finds passages by meaning, even when they do not share keywords. It handles paraphrases well but can miss exact codes and names.

Keyword Search

Matching exact terms (BM25), strong for names, codes and rare words.

Keyword search, often BM25, matches exact terms. It is strong for names, product codes and rare words that embeddings can blur.

Hybrid Search

Combining keyword and semantic scores for better recall.

Hybrid search combines keyword and semantic scores. It usually improves recall over either method alone, at the cost of more tuning.

Metadata Filtering

Restricting retrieval by attributes such as date, source, permission or product.

Filters restrict retrieval by attributes such as date, source, product or user permission. Permission filters are essential so users only retrieve documents they may read.

Retrieval Strategies

Top-k, multi-query, parent-document, query rewriting and filters tuned to the content.

Options include top-k retrieval, multi-query retrieval, parent-document retrieval and query rewriting. Choose based on the failure patterns you see in evaluation.

Reranking

A second model reorders retrieved candidates by relevance before generation.

A reranker, a second model, reorders retrieved candidates by relevance before generation. It improves precision at the cost of extra latency.

Context Construction

Selecting, ordering and formatting retrieved passages inside the prompt.

Context construction selects, orders and formats retrieved passages inside the prompt, with source labels. Include only what supports the answer, since extra text can distract the model.

Grounded Generation

Instructing the model to answer only from provided sources and say when they are insufficient.

The model is instructed to answer only from the provided sources and to say when they are insufficient. Test that it does this on questions the documents do not cover.

Citation and Source Attribution

Linking each claim to its source so users can verify.

Each claim links to its source so users can verify it. Citations build trust, but they must point to the passage that actually supports the claim.

RAG Evaluation

Measuring retrieval and answer quality separately on a labelled question set.

Evaluate retrieval and answer quality separately on a labelled set of questions. A wrong answer may come from retrieval or from generation, and you need to know which.

Retrieval Precision and Recall

Precision is the share of retrieved passages that are relevant; recall is the share of relevant passages retrieved.

Precision is the share of retrieved passages that are relevant. Recall is the share of relevant passages that were retrieved. Both must be measured to tune chunking and ranking.

Faithfulness and Context Relevance

Faithfulness checks the answer is supported by the context; context relevance checks the context fits the question.

Faithfulness checks the answer is supported by the context. Context relevance checks the retrieved context fits the question. Both can be scored with a model-based judge validated against human labels.

RAG Performance Optimisation

Tuning chunking, embeddings, hybrid search, reranking, caching and prompts for quality, speed and cost.

Tune chunking, embeddings, hybrid search, reranking, caching and prompts for quality, speed and cost. Change one factor at a time and re-run the evaluation.

Production RAG Monitoring

Tracking retrieval quality, unanswered questions, latency, cost and freshness in live use.

Monitor retrieval quality, unanswered questions, latency, cost and document freshness in live use. Questions with no good source are a useful signal for content gaps.

Summary of terms

Term	Meaning

RAG Fundamentals	Retrieve relevant documents at request time and give them to the model to ground its answer.
Why RAG Is Required	Models lack your private or recent data and can hallucinate; RAG supplies sources and enables citations.
RAG vs Fine-Tuning	RAG adds knowledge at query time and is easy to update; fine-tuning changes behaviour or style. They are often combined.
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Hybrid Search	Combining keyword and semantic scores for better recall.
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Retrieval Strategies	Top-k, multi-query, parent-document, query rewriting and filters tuned to the content.
Reranking	A second model reorders retrieved candidates by relevance before generation.
Context Construction	Selecting, ordering and formatting retrieved passages inside the prompt.
Grounded Generation	Instructing the model to answer only from provided sources and say when they are insufficient.
Citation and Source Attribution	Linking each claim to its source so users can verify.
RAG Evaluation	Measuring retrieval and answer quality separately on a labelled question set.
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Faithfulness and Context Relevance	Faithfulness checks the answer is supported by the context; context relevance checks the context fits the question.

RAG Performance Optimisation	Tuning chunking, embeddings, hybrid search, reranking, caching and prompts for quality, speed and cost.
Production RAG Monitoring	Tracking retrieval quality, unanswered questions, latency, cost and freshness in live use.

Practice exercise: Build a small RAG assistant over ten documents and measure retrieval precision and recall on twenty questions.

Self-check

1. Explain RAG Fundamentals in two sentences to a non-technical stakeholder, then say what you would measure for it.
2. Explain FAISS in two sentences to a non-technical stakeholder, then say what you would measure for it.
3. Explain Production RAG Monitoring in two sentences to a non-technical stakeholder, then say what you would measure for it.

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